

Week 7 Capstone Progress Documentation

PW26_MG_02

Week 7 marks one of the most defining and challenging phases of our application's journey. We have identified a **critical unsafe JSON deserialization vulnerability** deeply embedded within our request processing and data transformation layers. This discovery represents not just a technical issue, but a pivotal moment — a turning point where we choose resilience over risk and long-term integrity over short-term convenience.

This vulnerability exposes the system to severe threats, including remote code execution, unauthorized access, and potential full service compromise. Rather than applying temporary patches, we are committing to a fundamental architectural shift. This is not a minor fix — it is a complete redesign of how we define trust boundaries, validate input, and handle data across services.

Our remediation strategy involves eliminating unsafe parsing mechanisms, enforcing strict schema validation, redesigning deserialization workflows, and implementing controlled object mapping. Comprehensive regression testing, security validation, and penetration testing will ensure that every integration point is hardened. This transformation requires careful execution, and therefore remediation will continue until **7th March**, as security remains the highest priority for any application built to endure.

Until full remediation is achieved, we have implemented interim safeguards including enhanced monitoring, stricter request filtering, and reduced exposure of vulnerable endpoints. The system is being actively guarded while we rebuild it stronger.

This phase represents the most challenging shift in our development lifecycle — a conscious move toward security-first engineering. It reinforces a core principle: trust must be earned through design, not assumed through code. By confronting this vulnerability head-on, we are not just fixing a flaw — we are elevating the foundation of our entire system.